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4.4.1/5.4.1: Week 4/5 Homework - Thermodynamics, Circles, Mechanical Waves

Individual
AssignmentProgress: 9/58 Answered ▼ Score: _____
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▼ General Instructions



Make sure you answer all questions and select the "Submit for Grading" button when you are done to submit your work.

- The assignment will be auto-submitted on the due date. If you have not finished the assignment at that point, you will get credit for what you have completed.
- You may request the assignment be re-opened so that you can finish it after the deadline. **There is a 10% deduction per week late with a maximum deduction of 50%.**

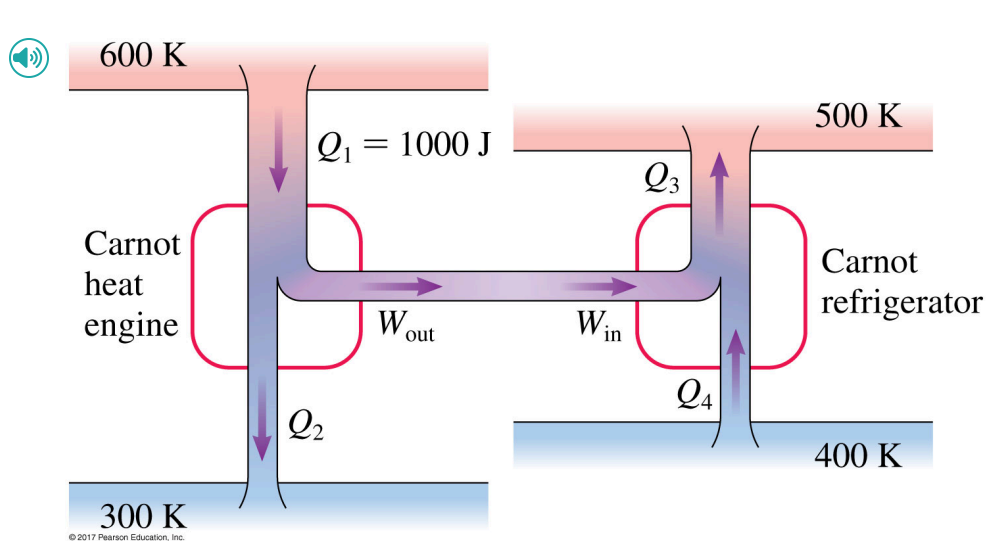
Please note the following:

- **Questions 1 - 17 are required.**
- **Question 18 is an optional reflection question.** It is not worth any points and there is nothing to submit; it is included to encourage you to go back and review the assignment, think about what you struggled with, and determine *why* you struggled so you can address the issues either on your own or during office hours.
- **Question 19 - 21 are optional; they provide more scaffolding if you are struggling with some of the concepts.**
 - Question 19 is an optional question to help you identify parameters in the sine function for waves.
 - Question 20 is a scaffolded version of Question 10. Please note *you must submit an answer to Question 10 for credit*; Question 20 is there to guide you through the multi-step process if you need or want the help, but it is not worth any points.
 - Question 21 is a scaffolded version of Question 13. Please note *you must submit an answer to Question 13 for credit*; Question 21 is there to guide you through the multi-step process if you need or want the help, but it is not worth any points.

You are always encouraged to do more problems for practice. You can find more relevant problems at the end of Chapter 21 and 16 of our textbook.

> Question 1: Carnot Cycle

- The diagram below shows a Carnot heat engine driving a Carnot refrigerator. Note that the reservoir temperatures are given in the diagram for both the engine and the refrigerator, as is the heat Q_1 .



1. What is Q_2 ? Report your answer in Joules and round to the nearest whole number.

Hint please?

J

Submit Answer

✓ 1 / 5 submissions used

Score: 1/1

Comments:



Correct!

2. What is the coefficient of performance for the refrigerator? Round your answer to the nearest whole number.

Hint please?

Submit Answer

✓ 1 / 5 submissions used

Score: 1/1

Comments:



Correct!

3. What is Q_4 ? Report your answer in J and round to the nearest whole number.



Why Q4 before Q3 ...?

J

Submit Answer

✓ 1 / 5 submissions used

Score: 1/1

Comments:



Correct!

4. What is Q_3 ? Report your answer in J and round to the nearest whole number.



Hint please?

J

Submit Answer

✓ 2 / 5 submissions used

Score: 1/1

Comments:



Correct!

> Question 2: Carnot Cycle

-  A Carnot heat engine operates between reservoirs at 182°C and 0°C . The engine extracts 25 J of energy from the hot reservoir per cycle.

-  1. How many cycles will it take the engine to lift a 10 kg mass a height of 10 m? Round your answer to the nearest whole number. 

Hint please?

cycles

Submit Answer

✓ 2 / 5 submissions used

Score: 1/1

Comments:



Correct!

> Question 3: Carnot Cycle

-  A typical coal-fired power plant burns 300 metric tons of coal *every hour* to generate 750 MW of electricity. Note the following:
- 1 metric ton = 1000 kg.
 - The density of coal is 1500 kg/m^3
 - The heat of combustion of coal is 28 MJ/kg
 - M = mega, which is 10^6

For this problem, assume that all heat is transferred from the fuel to the boiler and that all the work done in spinning the turbine is transformed into electric energy.



1. If the coal is piled up in a 10 m x 10 m room, how tall must the pile be to operate the power plant for 1 day? Report your answer in m and round to the nearest whole number.



Hint please?

m

Submit Answer

✓ 1 / 5 submissions used

Score: 1/1

Comments:



Correct! For perspective, that's a pile of coal that's about 157 feet tall in the 10 m x 10 m room.



2. What is the power plant's thermal efficiency? Report your answer with 2 digits after the decimal.



Treat the power plant as a heat engine.

Hint please?

Submit Answer

✓ 1 / 5 submissions used

Score: 1/1

Comments:

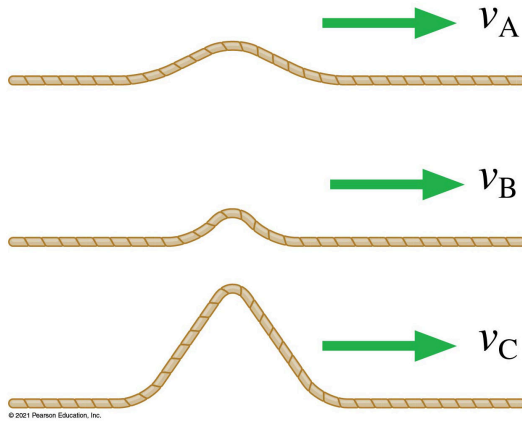


Correct!

> Question 4: Wave Speed/Wave on a String



The three wave pulses shown in the diagram below travel along **the same** stretched string.



1. How do the wave speeds, v_A , v_B , v_C , compare to each other? Why is this the case?



The speed of a wave pulse traveling along a string depends on the in the string and the of the string. It does not depend on the . Since all three pulses travel along , the and the are the same in all cases, which means .

Reset ↺

Submit Answer

✓ 1 / 5 submissions used

Score: 1/1

Comments:



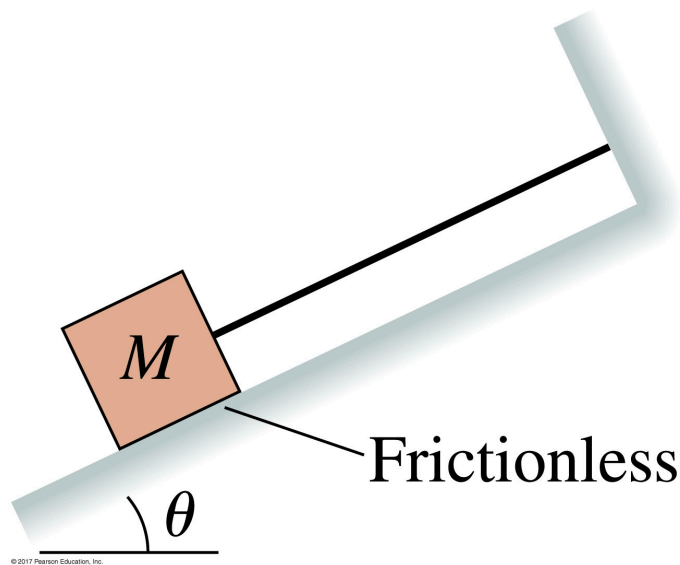
Correct!

▼ Question 5: Wave on a String



The string in the diagram below has a linear mass density of μ . Note that this is *not* the coefficient of friction!





1. Derive an expression for the speed of waves on the string in terms of given variables



M, g, θ, μ .

Don't just guess and put terms in the field below until you get it right. Actually do the physics to determine the answer.

Hide Hint



Hint:

- Draw a free body diagram for the block and use it to determine an expression for the tension force in terms of the relevant variables.
- Once you know F_T , use it in the equation for the speed of a wave pulse on a string.

$$v_{string} = \sqrt{\frac{M g \sin \theta}{\mu}}$$

Choices: **Reset** ↺

f_s

f_k

$\sin^2 \theta$

μ^2

$\cos \theta$

$\tan^2 \theta$

$\cos^2 \theta$

$\tan \theta$

Submit Answer

✓ 1 / 5 submissions used

Score: 1/1

Comments:



Correct!

▼ Question 6: Sinusoidal Waves



A sinusoidal wave has an angular frequency of 30 rad/s and a wavelength of 2.0 m.



1. What is the wavenumber for the wave?



Keep all quantities in base SI units. The follow-up question will ask you for the correct units for the wavenumber.

Hint please?

Submit Answer

0 / 5 submissions used

Score: 0/1



2. What are the units for the wavenumber of the wave you reported in the previous part of this problem?



☐ 1/m

☐ rad/m

☐ m/rad

☐ rad

☐ m/s

Submit Answer

0 / 4 submissions used

Score: 0/1

3. What is the wave speed for the wave?



Make sure all quantities are in base SI units. The follow-up question will ask you for the units for this value.

Hint please?

Submit Answer

0 / 5 submissions used

Score: 0/1

4. What are the units for the wave speed of the wave you reported in the previous part of this problem?



☐ m/rad

☐ rad/m

☐ m/s

☐ rad

☐ 1/m

Submit Answer

0 / 4 submissions used

Score: 0/1

> Question 7: Sinusoidal Waves



The displacement of a wave traveling in the negative y direction is given by the equation below, where y is in meters and t is in seconds.


$$D(y, t) = (5.2 \text{ cm}) \sin(5.5y + 72t)$$



1. What is the amplitude of the wave? Report your answer in cm.



Hint please?

cm

Submit Answer

0 / 5 submissions used

Score: 0/1



2. What is the frequency f of the wave? Report your answer in Hz, or 1/s.



Hint please?

Hz

Submit Answer

0 / 5 submissions used

Score: 0/1



3. What is the wavelength of the wave? Report your answer in meters, m.



Hint please?

m

Submit Answer

0 / 5 submissions used

Score: 0/1

4. What is the speed of the wave? Report your answer in m/s.



Hint please?

m/s

Submit Answer

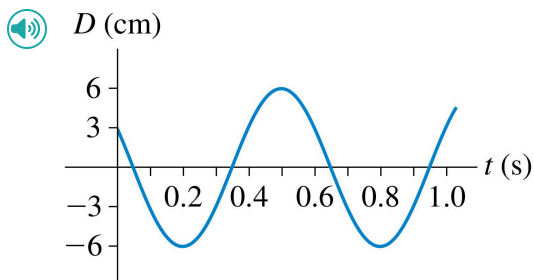
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Score: 0/1

> Question 8: Sinusoidal Waves

- The diagram below shows history graph of a sinusoidal wave; it shows the displacement of the wave as a function of time.

- The graph shows the displacement of the wave at each time that it passes through the $x = 0$ cm location.
- Note that the wave is traveling to the left at a speed of 2.0 m/s.



History graph at $x = 0$ m
Wave traveling left at 2.0 m/s

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1. What is the amplitude of this wave? Report your answer in cm.



cm

Submit Answer

0 / 5 submissions used

Score: 0/1



2. What is the frequency of this wave? Report your answer in Hz, or 1/s.



Hint please?

Hz

Submit Answer

0 / 5 submissions used

Score: 0/1



3. What is the wavelength of the wave? Report your answer in m.



Hint please?

m

Submit Answer

0 / 5 submissions used

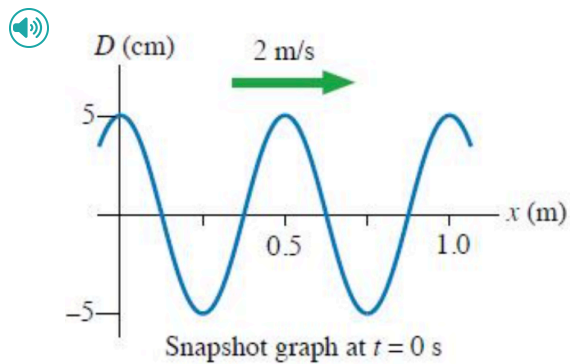
Score: 0/1

> Question 9: Sinusoidal Waves



The figure below shows a snapshot of a sinusoidal wave traveling to the right.





1. What is the wave amplitude? Express your answer in cm.

 cm

Submit Answer

0 / 5 submissions used

Score: 0/1

2. What is the wavelength of the wave? Express your answer in m.

 m

Submit Answer

0 / 5 submissions used

Score: 0/1

3. What is the wave number of the wave? Express your answer in rad/m.



Hint please?

 rad/m

Submit Answer

0 / 5 submissions used

Score: 0/1

4. What is the frequency for this wave? Express your answer in Hz, or inverse seconds.



Hint please?

Hz

Submit Answer

0 / 5 submissions used

Score: 0/1

5. What is the oscillation period for the wave? Express your answer in s.



Hint please?

s

Submit Answer

0 / 5 submissions used

Score: 0/1

6. What is the angular frequency of the wave? Express your answer in rad/s.



Hint please?

rad/s

Submit Answer

0 / 5 submissions used

Score: 0/1

> Question 10: Sinusoidal Waves and Wave Speed

A scaffolded version of this question is provided at the end of the assignment, as Question 20. **This is the version that is required for the assignment;** if you are struggling with the thought process, see the

optional scaffolded version to help guide you through it, but don't forget to come back and enter your answers in this question to earn points on the assignment!

A string that is under 50.0 N of tension has a linear density of 5.0 g/m. A sinusoidal wave with an amplitude of 3.0 cm and wavelength of 2.0 m travels along this string.



1. What is the maximum speed of a **particle** on the string? Report your answer with 2 digits after the decimal in m/s.



Be careful, this is asking about the maximum speed of a particle, not the maximum speed of the wave pulse! They can be related, but it takes a few steps. See the scaffolded version of this question at the end of the assignment (Question 20) for help and hints!

Hint please?

m/s

Submit Answer

0 / 5 submissions used

Score: 0/1

> Question 11: Sinusoidal Waves and Wave on a String



A sinusoidal wave travels along a stretched string. A particle on the string has a maximum speed of 2.0 m/s and a maximum acceleration of 200 m/s^2 .



1. What is the frequency of the wave, f ? Report your answer in Hz and round to the nearest whole number.



Hint please?

Hz

Submit Answer

0 / 5 submissions used

Score: 0/1



2. What is the amplitude of the wave, A? Report your answer in cm and round to the nearest whole number.



Hint please?

cm

Submit Answer

0 / 5 submissions used

Score: 0/1

> Question 12: Power and Intensity



Sound is detected when a sound wave causes the eardrum to vibrate. Typically, the diameter of the eardrum is about 8.4 mm in humans. When someone speaks to you in a normal tone of voice, the sound intensity at your ear is approximately $1.0 \times 10^{-6} \text{ W/m}^2$.



1. How much energy is delivered to your eardrum each second? Report your answer in pJ, where 1 pico = 1×10^{-12} (... your answer should be a pretty small number).



Hint please?

pJ

Submit Answer

0 / 5 submissions used

Score: 0/1

> Question 13: Power, Intensity, and Decibels



A scaffolded version of this question is provided at the end of the assignment, as Question 21. **This is the version that is required for the assignment;** if you are struggling with the thought process, see the optional scaffolded version to help guide you through it, but don't forget to come back and enter your answers in this question to earn points on the assignment!

The sound intensity level 5.0 m away from a large power saw is 100 dB.



1. Determine the distance at which the sound intensity level is a more tolerable 80 dB.



Report your answer in meters and round to the nearest whole number.

m

Submit Answer

0 / 5 submissions used

Score: 0/1

> Question 14: Power, Intensity, and Decibels



30 seconds of exposure to 115 dB sound can damage your hearing, but a much quieter 94 dB may begin to cause damage after 1 hour of continuous exposure. You are going to an outdoor concert and you'll be standing near a speaker that emits 50 W of acoustic power as a spherical wave.



1. What minimum distance should you be from the speaker to keep the sound intensity level below 94 dB? Report your answer in m and round to the nearest whole number.



Hint please?

m

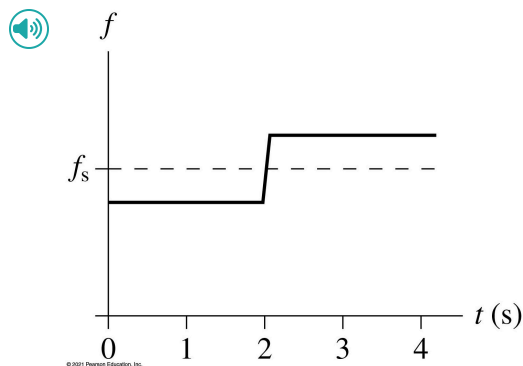
Submit Answer

0 / 5 submissions used

Score: 0/1

> Question 15: Doppler Effect

-  You stand at $x = 0$ m listening to a sound that is emitted at frequency f_s . The diagram below shows the frequency you hear during a four second interval.



-  1. During the first 2 seconds, is the source moving toward you, away from you, or is it not moving? How do you know?

During the first two seconds, the sound source is .

The graph shows that the frequency of the source is f_s , the original frequency for the non-moving case. If the frequency you hear is the original frequency, it means the source is since it takes for each sound wave to reach your ears.

Submit Answer

0 / 5 submissions used

Score: 0/1

2. What does the sound source do at $t = 2$ s? How do you know?

The sound source at $t = 2$ s.

We know this because the frequency that we hear , going from something the original source frequency f_s to something the original source frequency f_s . Whether the frequency we hear is greater or less than the original source frequency tells us how the object is moving, so if that changes, the direction the object is moving .

Submit Answer

0 / 5 submissions used

Score: 0/1



3. From 2 - 4 s, is the source moving toward you, away from you, or is it not moving? How do you know?



From 2 - 4 s, the sound source is .

The graph shows that the frequency of the source is f_s , the original frequency for the non-moving case. If the frequency you hear is the original frequency, it means the source is since it takes for each sound wave to reach your ears.

Submit Answer

0 / 5 submissions used

Score: 0/1

> Question 16: Doppler Effect



A friend of yours is loudly singing a single note at 440 Hz while racing toward you at 25.0 m/s on a day when the speed of sound is 340 m/s.



1. You are standing stationary while this happens. What frequency of sound do you hear?



Report your answer in Hz with 2 digits after the decimal.

Hint please?

Hz

Submit Answer

0 / 5 submissions used

Score: 0/1



2. You are still standing stationary, but you decide to sing back at your friend as they continue to approach you. You sing at 440 Hz; what frequency does your friend hear?



Report your answer in Hz with 2 digits after the decimal.

Hint please?

Hz

Submit Answer

0 / 5 submissions used

Score: 0/1

> Question 17: Doppler Effect



A physics professor demonstrates the Doppler effect by tying a 600 Hz sound generator to a 1.0 m long rope and whirling it around their head in a horizontal circle at 100 rpm. Take the speed of sound

in air to be 343 m/s.

The goal of this problem is to determine the maximum and minimum sound frequencies a student hears in this situation.



1. Before doing anything with the Doppler effect, you will need to know the speed of the sound generator (the source) in m/s. Convert the given angular speed to linear speed and enter your answer below with 2 digits after the decimal.



Hint please?

m/s

Submit Answer

0 / 5 submissions used

Score: 0/1



2. Now that you know the speed of the source, determine the **maximum** frequency a student hears for this demonstration. Report your answer in Hz and round to the nearest whole number.



Hint please?

Hz

Submit Answer

0 / 5 submissions used

Score: 0/1



3. Now that you know the speed of the source, determine the **minimum** frequency a student hears for this demonstration. Report your answer in Hz and round to the nearest whole number.



Hint please?

Hz

Submit Answer

0 / 5 submissions used

Score: 0/1

> Question 18: Revisit/Reflect on Any Question 1 - 17 (Optional)



The goal of this question is for you to go back to one of the problems in the assignment and look at your work/think about the actual problem-solving process for that problem. **This question is not worth any points and there is nothing to submit.**

Why am I asking you to do this? Isn't it enough to solve the problem and get the answer correct?

- You're right, it *is* a good thing to get the answer correct, but what's more important - especially in terms of learning and being able to apply your learning to the exam - is understanding the process used to solve complicated problems.
- It's fine if you struggle with the process - that's what learning is all about - but it won't do you any good on the exam if you never go back and think about that process.



1. Pick at least one of the questions from this assignment to revisit/reflect on.



- What part of the problem-solving process was the most confusing or challenging? Why was this the case?
- What part of the problem-solving process made the most sense to you? Why was this the case?

> Question 19: Mathematics of a Traveling Wave (OPTIONAL)



This question is designed to help solidify variables. It is not required, but if you are struggling with identifying terms and keeping track of which variable is which, this question would be good practice.

One formula for the vertical displacement of a wave - for example, a string - traveling in the x direction, is given below. The questions in this problem refer to this equation.

 $y(x, t) = A \sin(kx - \omega t)$



 1. Which of the quantities in the equation are *independent* variables? Choose all that apply.



Select all that apply:

☐ k ☐ ω ☐ A ☐ t ☐ x

Submit Answer

0 / ∞ submissions used

Heads Up! You need to submit this answer for it to save and be graded.

2.



This question is hidden until you answer the previous question.

3.



This question is hidden until you answer the previous question.

4.



This question is hidden until you answer the previous question.

5.



This question is hidden until you answer the previous question.

6.



This question is hidden until you answer the previous question.

> Question 20: Sinusoidal Waves and Wave Speed (Scaffolded Version - OPTIONAL)



This is an optional scaffolded version of Question 10. This version is not worth any points, but will help guide you through to the final answer for the required Question 10 if you need or want the extra help.

If you do this version of the problem, make sure you also go back and input your final answer into Question 10. Doing this question will not earn you any points.

A string that is under 50.0 N of tension has a linear density of 5.0 g/m. A sinusoidal wave with an amplitude of 3.0 cm and wavelength of 2.0 m travels along this string.

The end goal of this problem is to determine the maximum speed of a *particle* on the string.



1. What is the equation to determine the speed of the wave pulse on a string? There may be more than one correct answer.



Select all that apply:



☐
$$v = \frac{\Delta x}{\Delta t}$$

☐
$$v = A\omega$$

☐
$$v = \lambda f$$

☐
$$v = \sqrt{\frac{T}{\mu}}$$

Submit Answer

0 / ∞ submissions used

Heads Up! You need to submit this answer for it to save and be graded.



2. What is the equation to determine the speed a particle on the wave?





☐
$$v = \frac{\Delta x}{\Delta t}$$

☐
$$v = \lambda f$$

☐
$$v = \sqrt{\frac{T}{\mu}}$$

☐
$$v = A\omega$$

Submit Answer

0 / ∞ submissions used

3.

This question is hidden until you answer the previous question.



4.

This question is hidden until you answer the previous question.



5.

This question is hidden until you answer the previous question.



➤ Question 21: Power, Intensity, and Decibels (Scaffolded Version - OPTIONAL)



This is an optional scaffolded version of Question 13. This version is not worth any points, but will help guide you through to the final answer for the required Question 13 if you need or want the extra help.

If you do this version of the problem, make sure you also go back and input your final answer into Question 13. Doing this question will not earn you any points.

The sound intensity level 5.0 m away from a large power saw is 100 dB. **The goal of this problem is to determine the distance at which the sound intensity level is a more tolerable 80 dB.**



There are a couple of different ways to approach this problem. We will work through the method that is a little more straightforward and involves less algebra with the log function, though the downside is that it involves more steps. At the end of this problem, the other approach is briefly outlined (fewer overall steps, but you have to do more algebra with the log function).



1. The end goal of this problem is to find a distance, r , away from the power saw. Which equation will let us do this?



☐ $\beta = (10 \text{ dB}) \log_{10} \left(\frac{I}{1.0 \times 10^{-12} \text{ W/m}^2} \right)$

☐ $f_- = \frac{f_0}{1 + \frac{v_s}{v}}$

☐ $I = \frac{P}{4\pi r^2}$

☐ $f_+ = \frac{f_0}{1 - \frac{v_s}{v}}$

Submit Answer

0 / ∞ submissions used

2.



This question is hidden until you answer the previous question.

3.



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4.



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